# Reading list, organised by a reviewer's three tiers

The earlier reading map was organised by week. This one is organised by what a PI said they look for, so you can see what to read for each competency, which module part already covers it, and which problems test it. Where a book or paper is named, the title and author are recalled from memory and the edition may have moved — **check each against a library catalogue before buying**. Chapter numbers differ between editions, which is why the topic is named rather than a number given.

### How to use it

Read the module chapter first to know what matters, then the book section for the mechanism and figures, then work the problems. The module parts are the files listed on the tracker's Materials tab; problem codes are in its Problems tab.

## Tier 1 — the non-negotiable core

| Competency | Read | Module | Problems |
|---|---|---|---|
| **Central carbon metabolism and redox balancing** | Lehninger, *Principles of Biochemistry* — glycolysis, TCA, pentose phosphate, oxidative phosphorylation. Nelson & Cox, any recent edition.<br>Stephanopoulos, Aristidou & Nielsen, *Metabolic Engineering: Principles and Methodologies* — the stoichiometry and yield chapters, which is the book this tier is built on. | Part 14 ch B4–B7 (computed); part 2 ch 18–19; part 15 X1–X4 | A2, K1, K3, W2, W3, X1, X2, X3, EV3 |
| **Theoretical yields and carbon recovery** | Stephanopoulos ch 2–3 again, worked rather than read. For the thermodynamic layer, Noor et al. on max-min driving force (recalled — search "maximum driving force pathway thermodynamics"). | Part 2 ch 14, 18; part 14 ch B7 | A2, X1, X2, PP2, EV3 |
| **CRISPR: Cas9, Cas12a, base editors, prime editing** | Doudna & Charpentier's review of CRISPR-Cas9 genome editing (Science, 2014) for the foundation; a recent review of base and prime editing for the mechanisms — this field moves fastest, so prefer a review under two years old. For bacteria specifically: a review of CRISPR-assisted recombineering. | **Part 21 ch W1**; part 7 ch 23–24 | **EV1, EV2**, PP5, PP6 |
| **Recombineering, markerless editing, integration** | Datsenko & Wanner (PNAS, 2000) — the one-step inactivation paper, still the template for how this is done. Court, Sawitzke & Thomason on recombineering (Annu. Rev. Genet.). | Part 7 ch 23; part 18 PP6 | PP6, EV1 |
| **Enzyme kinetics** | Cornish-Bowden, *Fundamentals of Enzyme Kinetics* — steady state, inhibition, two-substrate. Bar-Even et al., "The moderately efficient enzyme" (Biochemistry, 2011) for what real $k_{cat}$ and $K_M$ values look like. | Part 14 ch B3; chapter S3; part 2 ch 12 | B1, X5, X6, X8, L2 |
| **Directed evolution and rational design** | Arnold's Nobel lecture or a recent directed-evolution review for the strategy; Romero & Arnold (Nat. Rev. Mol. Cell Biol., 2009) on exploring protein fitness landscapes. | Part 8 ch 29–30 | L1, L2, Y5 |
| **Bioprocess: $k_La$, OUR, fed-batch, burden** | Doran, *Bioprocess Engineering Principles* — mass transfer, mixing, heat transfer, worked as problems. Shuler, Kargi & DeLisa for the fermentation chapters.<br>Scott et al. (Science, 2010) and Basan et al. (Nature, 2015) for the growth laws and overflow metabolism. | Part 11 ch 44; part 19 ch T4; part 14 ch B9; part 20 ch V1 | D1–D6, X4, X9–X12, PP9, MP1–MP3 |

## Tier 2 — the accelerator skills

| Competency | Read | Module | Problems |
|---|---|---|---|

| | | | |
|---|---|---|---|
| **FBA and genome-scale models** | Palsson, *Systems Biology: Constraint-based Reconstruction and Analysis* — the standard text. Orth, Thiele & Palsson, "What is flux balance analysis?" (Nat. Biotechnol., 2010) — read this one first, it is short. Then the COBRApy documentation, worked through on the *E. coli* core model. | Part 10 ch 35–38; part 17 MC1–MC5 | E1, E2, MC1–MC5, Y1 |
| **Model curation** | Thiele & Palsson's protocol for generating a metabolic reconstruction (Nat. Protoc., 2010); the MEMOTE paper for automated testing. | Part 10 ch 37; part 17 ch MC-series | MC1–MC5 |
| **Multi-omics literacy** | A recent review of multi-omics integration in metabolic engineering. For the statistics that make or break it: Love, Huber & Anders on DESeq2 (Genome Biol., 2014) for count data. | Part 4 ch 51; part 9 ch 33; chapter S5 | Y2, Z5, Z6, Z9, R4 |
| **$^{13}$C flux analysis** | Wiechert on $^{13}$C metabolic flux analysis (Metab. Eng., 2001); Antoniewicz's reviews for the practical side and for natural-abundance correction. | Part 11 ch 43; chapter S5 | C1, C2, C3, R4 |
| **ML for protein and pathway design** | Jumper et al. on AlphaFold (Nature, 2021) — read the results and the confidence metrics, not the architecture. A recent review of protein language models and variant-effect prediction. For the campaign side, a review of ML-guided directed evolution. | Part 8 ch 29–30; part 11 ch 41–42 | P1, P2, Y5 |
| **Biosensors and high-throughput screening** | A review of transcription-factor biosensors for metabolic engineering; Zhang, Jensen & Keasling's work on sensor-driven screening (recalled). | Part 8 ch 26–27 | I1, O1, L1 |
| **FACS and droplet microfluidics** | A methods review on droplet microfluidic screening for enzymes and strains; for FACS, any flow-cytometry handbook's chapters on gating and sort purity. | **Part 21 ch W3** | **EV5, EV6** |

# Tier 3 — the specialised edge

| Competency | Read | Module | Problems |
|---|---|---|---|
| **Statistical design of experiments** | Box, Hunter & Hunter, *Statistics for Experimenters* — factorials, aliasing, steepest ascent. Myers, Montgomery & Anderson-Cook, *Response Surface Methodology* — canonical analysis and design comparison. | **Part 20 ch V4**; part 19 ch T3; part 2 ch 17 | U1, U2, F1, F2, X13–X16, DE1–DE4 |
| **Python automation and NGS pipelines** | VanderPlas, *Python Data Science Handbook*, free online, for numpy, pandas and plotting. The breseq documentation for variant calling, worked on a real SRA dataset. | Part 9 ch 32, 34; part 10 ch 36 | J1, Y2, Y6, R1–R5 |
| **Non-conventional chassis** | A review of non-model industrial hosts; for autotrophs and C1 substrates, a review of gas fermentation and of methanol bioconversion. Part 6's host table is the summary to argue from. | Part 6 ch 58–61; part 15 X4 | K3, K4, N2, W1, X4, PP1 |
| **Techno-economic assessment** | Humbird et al.'s NREL design report for a cellulosic process (2011) — the worked example everyone cites; a recent TEA review for bio-based chemicals. *See the note below: this belongs higher than Tier 3.* | Part 4 ch 49; part 15 X22 | G1, D5, T2, X22, PP12 |
| **Life cycle analysis** | ISO 14040 and 14044 for the framework; the ecoinvent or GREET documentation for emission factors; one published LCA of a bio-based chemical to see a real system boundary. | **Part 21 ch W4** | **EV7, EV8** |

### Which file is which "part"

The problem supplements are titled as parts inside the PDF but named by their problem range in the filename. The mapping: **Part 15** = `me-module-supplement-problems-X1-X24.pdf` (X1–X24, Y1–Y6) · **Part 16** = `me-module-supplement-problems-Z1-Z9.pdf` (Z1–Z9, separations, quantitative omics, consortia) · **Part 17** = `me-module-supplement-problems-MC1-R5.pdf` (MC1–MC5, R1–R5) · **Part 18** = `me-module-supplement-problems-PP1-PP12.pdf` (PP1–PP12). Parts 19 to 23 carry their part number in the filename.

### Also in Tier 3 and the product classes — Part 16

| Competency | Read | Module | Problems |
|---|---|---|---|
| **Downstream separations as unit operations** | Harrison, Todd, Rudge & Petrides, *Bioseparations Science and Engineering* — chromatography, membranes, disruption; a vendor process-development handbook for breakthrough-curve protocols. | Chapter S1; part 19 ch T5; **part 16** | **Z1 Z2 Z3 Z4**, PP10, PP11, DS3 |
| **Quantitative metabolomics and proteomics** | A review of absolute protein quantification (AQUA and related); a methods review on metabolite quenching and extraction. | Chapter S5; **part 16** | **Z5 Z6 Z9**, R4 |
| **Microbial consortia** | A review of synthetic microbial consortia for metabolic engineering, read for the stability arguments rather than the examples. | Chapter S4; **part 16** | **Z7 Z8** |

## Product-class reading, for the four named classes

| Class | Read | Module | Problems |
|---|---|---|---|
| **Terpenoids** | Paddon et al. on semi-synthetic artemisinin (Nature, 2013) — the canonical end-to-end paper, worth reading twice; a review of isoprenoid pathway engineering. | Part 5 ch 53; part 15 X3 | Q1, Q2, S1–S5, X3 |
| **Flavonoids and polyphenols** | A review of microbial flavonoid production, read specifically for what each group did about malonyl-CoA; a review of P450 functional expression in microbial hosts. | **Part 21 ch W2** | **EV3, EV4** |
| **Tailored biopolymers** | A review of PHA production and of the recovery routes; Choi & Lee on PHA economics (recalled). | Part 5 ch 56; part 18 PP10 | K1, K2, PP10 |
| **Rare amino acids** | Becker & Wittmann on systems metabolic engineering of *C. glutamicum* for lysine (recalled); a review of amino-acid fermentation. | Part 6 ch 59; part 18 PP1 | W2, PP1 |
| **Therapeutic proteins** | Baneyx & Mujacic on folding and inclusion bodies (Nat. Biotechnol., 2004); a review of secretory production in *K. phaffii*; a glycoengineering review for humanised yeast. | **Part 19 ch T1–T2** | **PR1–PR8** |

## The six-month roadmap, mapped onto this module

| Phase | The reviewer's focus | What to actually do here |
|---|---|---|
| **Months 1–2** | Advanced systems and flux analysis | Part 14 chapters B4–B7 and part 2 chapters 13–19. Work A1, A2, X1–X3, EV3. Then part 10 and the COBRApy tutorial, finishing with Y1 on the real BiGG model and MC1–MC5 on curation. |
| **Months 3–4** | Precision editing and chassis design | Part 7 chapters 20–25 and **part 21 chapter W1**. Work PP5, PP6, EV1, EV2. Then part 8 chapters 26–28 for promoter and RBS tuning, with PP3, PP4 and H1, H2. Add part 6 for the host table and PP1. |
| **Months 5–6** | Computational biology and omics | Part 9 and part 11 chapters 41–43, with chapter S5. Work P1, P2, Y2, Y5, Z5, Z6, Z9 and the instrument problems R1–R5 on the supplied data. Finish with part 4 chapter 49, **part 21 chapter W4**, and X22, PP12, EV7, EV8 — so the last thing you do is cost a process in dollars and in carbon. |

**One deliberate departure.** The reviewer places techno-economic assessment in Tier 3 as "easily learned in the first year". Problem G1 and X22 suggest otherwise: the titer–separation relationship decides whether a project is viable at all, and the input that dominates the uncertainty is one a laboratory does not control. A candidate who can say why a strain cannot reach a viable price at any plausible yield is making an argument that first-year catch-up does not replace. This list therefore keeps TEA in Tier 3 where the reviewer put it, and the roadmap schedules it anyway.

**On the citations.** Author names, journals and years above are recalled from memory and not verified — this environment has no network access. Treat each as a search string rather than a citation: look it up, confirm it exists and is the paper I meant, and replace the entry with the real one. Sheet 9 of the lookup workbook has fields for recording what you actually read.